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Exploratory Data Analytics (E1CSA231)

Micro Project Report

On

Global Unemployment Trend Analysis

Submitted by

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Table of Contents

Abstract

1. Introduction

    1.1 Project Overview

    1.2 Objectives of the Project

    1.3 Problem Statement

    1.4 Scope of the Project

2. Dataset Description

    2.1 Source of Data

    2.2 Dataset Overview

    2.3 Features

3. Data Preprocessing

4. Exploratory Data Analysis

5. Results and Insights

6. Conclusion

7. Future Scope

References

Appendix

Abstract

This project explores unemployment data across multiple countries from 1998 to 2022. The dataset was sourced from international labor statistics in Excel format. The purpose of this analysis is to identify unemployment trends, compare countries, and highlight year-on-year changes. Tools such as Python, Pandas, NumPy, Matplotlib, and Seaborn were used for preprocessing and visualization. Key findings show that unemployment rates vary widely across regions, with developing countries experiencing higher volatility compared to developed nations. For instance, Niger and Chad show sharp fluctuations, while Japan and Switzerland remain relatively stable.

1. Introduction

1.1 Project Overview

The project focuses on analyzing unemployment rates across countries over time. By studying yearly unemployment percentages, we aim to understand economic stability, labor market conditions, and regional disparities.

1.2 Objectives of the Project

- To analyze unemployment trends across countries.
- To identify anomalies and outliers in unemployment rates.
- To compare developed vs developing nations.
- To derive insights useful for policymakers and economists.

1.3 Problem Statement

Unemployment is a critical indicator of economic health. High unemployment rates can lead to poverty, instability, and reduced growth.

Problem: What are the global unemployment trends, and which countries face the highest volatility?

1.4 Scope of the Project

- Includes: Exploratory analysis of unemployment dataset (1998–2022).
- Excludes: Predictive modeling or forecasting.

2 Dataset Description

2.1 Source of Data

- Dataset provided in Excel format (Unemployment Data set.xlsx).
- Compiled from international labor statistics.

2.2 Dataset Overview

- Rows: ~150 countries
- Columns: Yearly unemployment rates (1998–2022), country codes, averages, max values.
- Data types: Numerical (percentages), categorical (country names/codes).

2.3 Features

Column Name	**Description**
Region	Name of the region
Date	Time period of data
Estimated Unemployment Rate	Percentage of unemployment
Estimated Employed	Number of employed individuals
Estimated Labour Participation Rate	Participation rate in workforce

3 Data Preprocessing

3.1 Data Cleaning

- Removed duplicate entries
- Checked for inconsistent data
- Ensured proper formatting of columns

3.2 Handling Missing Values

- Identified missing values in the dataset
- Missing values were either removed or filled using mean/median methods

3.3 Data Transformation

- Converted date column into proper datetime format
- Renamed columns for better readability
- Structured data for easier analysis

4 Exploratory Data Analysis

4.1 Univariate Analysis

- Analyzed distribution of unemployment rate
- Used histogram and bar charts
- Calculated mean and median values

4.2 Bivariate Analysis

- Compared unemployment rate across regions
- Analyzed relationship between employment and unemployment
- Used scatter plots and bar charts

4.3 Multivariate Analysis

- Analyzed multiple variables like region, time, and unemployment rate
- Used heatmaps and pair plots

4.4 Data Visualization

The following visualizations were used:

- Line plot for trend analysis
- Histogram for unemployment rate
- Bar chart for regional comparison

4.5 Correlation Analysis

- Used correlation matrix to identify relationships
- Observed correlation between employment and unemployment
- Visualized using heatmap

5 Results and Insights

- Unemployment rates vary significantly across regions
- Certain regions consistently show higher unemployment
- There are noticeable fluctuations in unemployment over time
- Higher labour participation does not always mean lower unemployment
- Economic and regional factors play a major role

Example Insight:

Regions with higher employment numbers tend to have relatively stable unemployment rates.

6 Conclusion

This project successfully analyzed unemployment trends using exploratory data analysis techniques. The analysis helped in understanding patterns, variations, and relationships in the dataset. The findings highlight the importance of regional and temporal factors in unemployment. Overall, the project provides valuable insights into unemployment trends.

7 Future Scope

- Apply machine learning models to predict unemployment
- Use a larger dataset for more accurate analysis
- Include additional economic factors
- Perform deeper statistical analysis

References

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Appendix

Python code:

```
 import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_excel("Unemployment Data set.xlsx")
#Global Average Unemployment Trend (1998–2022)
years = df.loc[:, '1998':'2022'].mean()
plt.figure()
years.plot()
plt.title("Global Average Unemployment Trend (1998–2022)")
plt.xlabel("Year")
plt.ylabel("Unemployment Rate")
plt.show()
#Unemployment Trend for Top 5 Countries
top5 = df.head(5)
plt.figure()
for i in range(5):
    plt.plot(top5.loc[i, '1998':'2022'], label=top5['Country Name'][i])
plt.xticks(rotation=45)
plt.title("Unemployment Trend for Top 5 Countries")
plt.xlabel("Year")
plt.ylabel("Unemployment Rate")
plt.legend()
plt.show()
#Unemployment Trend for One Country (Example: Qatar)
country = df[df['Country Name']=="Qatar"].iloc[:,2:27].T
```

```
plt.figure()
plt.plot(country)
plt.xticks(rotation=45)
plt.title("Unemployment Trend for Qatar")
plt.xlabel("Year")
plt.ylabel("Unemployment Rate")
plt.show()
#Top 10 Countries with Highest Average Unemployment
top10 = df.sort_values("Average World", ascending=False).head(10)
plt.figure()
plt.bar(top10['Country Name'], top10['Average World'])
plt.xticks(rotation=45)
plt.title("Top 10 Countries with Highest Average Unemployment")
plt.xlabel("Country")
plt.ylabel("Average Unemployment Rate")
plt.show()
#Top 10 Countries with Lowest Average Unemployment
low10 = df.sort_values("Average World").head(10)
plt.figure()
plt.bar(low10['Country Name'], low10['Average World'])
plt.xticks(rotation=45)
plt.title("Top 10 Countries with Lowest Unemployment")
plt.xlabel("Country")
plt.ylabel("Average Unemployment Rate")
plt.show()
```

```
#Average Unemployment by Country
plt.figure()
plt.hist(df['Average World'], bins=20)
plt.title("Distribution of Average Unemployment by Country")
plt.xlabel("Average Unemployment Rate")
plt.ylabel("Frequency")
plt.show()
#YoY Change Distribution
plt.figure()
plt.hist(df['YoY change'], bins=20)
plt.title("Year-on-Year Unemployment Change Distribution")
plt.xlabel("YoY Change")
plt.ylabel("Frequency")
plt.show()
#Countries with Highest YoY Change
top_change = df.sort_values("YoY change", ascending=False).head(10)
plt.figure()
plt.bar(top_change['Country Name'], top_change['YoY change'])
plt.xticks(rotation=45)
plt.title("Countries with Highest YoY Unemployment Change")
plt.xlabel("Country")
plt.ylabel("YoY Change")
plt.show()
#Countries with Lowest YoY Change
low_change = df.sort_values("YoY change").head(10)
plt.figure()
plt.bar(low_change['Country Name'], low_change['YoY change'])
```

```
plt.xticks(rotation=45)
plt.title("Countries with Lowest YoY Unemployment Change")
plt.xlabel("Country")
plt.ylabel("YoY Change")
plt.show()
#Average vs Maximum Unemployment
plt.figure()
plt.scatter(df['Average World'], df['Max World'])
plt.title("Average vs Maximum Unemployment Rate")
plt.xlabel("Average Unemployment")
plt.ylabel("Maximum Unemployment")
plt.show()
#YoY Change vs Average Unemployment
plt.figure()
plt.scatter(df['YoY change'], df['Average World'])
plt.title("YoY Change vs Average Unemployment")
plt.xlabel("YoY Change")
plt.ylabel("Average Unemployment")
plt.show()
#Heatmap of Correlation
plt.figure()
sns.heatmap(df[['Average World','Max World','YoY change']].corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

Link: https://colab.research.google.com/drive/1fgLZ-r4SaNB-vEvyWS30q6kuNolE3ea3?usp=sharing